Summary in Graph

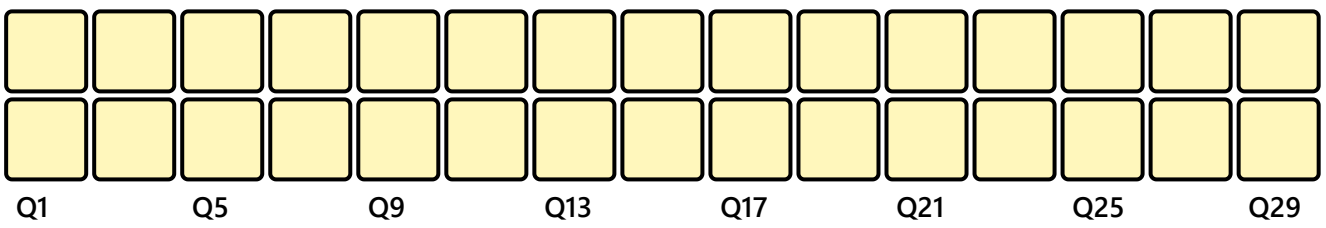
Exam Summary (GATE Overflow CS Test Series 2026 | Algorithms | Test 2)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (4) out of 38 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Algorithms

In an undirected graph with all edge weights equal, the shortest path from a node s to every other node is computed most efficiently, in terms of time complexity by _____.

A. Dijkstra's algorithm with s as the start node
B. Bellman-Ford algorithm with s as the start node
C. Performing a DFS starting from s

D. Performing a BFS starting from s

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following statements regarding Minimum Spanning Tree (MST) is/are TRUE? (Mark all the appropriate choices)

A. If e is the maximum weight edge in a connected graph G, then there is always a MST of G that does not include e
B. If all the edge weights of a graph G are positive, then any subset of edges that connects all the vertices and has minimum total weight forms a tree
C. If we decrease the weight of an edge in MST, it still remains a MST
D. If all the edge weights of a graph are increased by a value k, the MST remains the same

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following statements regarding DFS traversal is/are true? (Mark all the appropriate choices)

A. In a DFS in an undirected graph G, every edge is a tree edge or a back edge
B. In a DFS presence of a cross edge indicates the existence of a cycle
C. In a DFS of a graph G, if a node is visited twice, a cycle exists in G
D. In a DFS presence of a back edge indicates the existence of a cycle

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Let $G(V, E)$ be an undirected graph with some edge weights being negative but without any negative weight cycle. Which of the following is/are TRUE when G is given as input with a given source s? (Mark all the appropriate choices)

A. Dijkstra's algorithm will always correctly output the shortest path to all vertices from s
B. Dijkstra's algorithm will always fail to correctly output the shortest path to all vertices from s
C. Bellman-Ford algorithm will always correctly output the shortest path to all vertices from s
D. Bellman-Ford algorithm will always fail to correctly output the shortest path to all vertices from s

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which of the following algorithm can be used to efficiently calculate single source shortest paths in a Directed Acyclic Graph where all edge weights are equal?

A. Dijkstra's algorithm
B. Bellman-Ford algorithm
C. Topological Sort
D. Prim's algorithm

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Given a directed graph where weight of every edge is same, we can efficiently find the shortest path from a given source to destination using ____

A. Breadth First Traversal
B. Dijkstra's Shortest Path Algorithm
C. Bellman-Ford Algorithm
D. Depth First Search

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following is/are FALSE regarding Spanning Trees? (Mark all the appropriate choices)

A. A minimum spanning tree does not have any cycles
B. A spanning tree will have $n-1$ edges if the graph has n vertices
C. If an edge e belonging to a cut of the graph has weight smaller than any other edge in the same cut, then the edge e is present in all the minimum spanning trees of the graph
D. Removing any one edge from the spanning tree will make the original graph disconnected

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Given any two vertices in an undirected finite graph, which of the two graph traversal algorithms -- BFS and DFS -- can be used to find if there is path between them?

A. Only BFS
B. Both BFS and DFS
C. Only DFS
D. Neither BFS nor DFS

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following is/are FALSE in the case of a spanning tree T of a connected graph G?

A. T is never unique
B. T is a subgraph of G
C. T includes every vertex of G
D. T does not exist if G has a cycle

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Let G be a graph with n vertices and m edges. What is the tightest upper bound on the running time on Depth First Search of G? (Mark all the correct options)

A. $\Theta(m+n)$ if G is represented using adjacency matrix
B. $\Theta(m+n)$ if G is represented using adjacency list
C. $\Theta(n^2)$ if G is represented using adjacency matrix
D. $\Theta(n^2)$ if G is represented using adjacency list

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

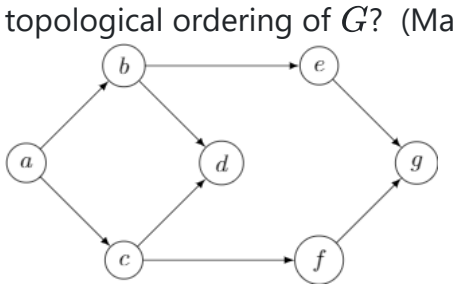
Q #11 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

If the worst case time complexity by the best algorithm for finding the outdegrees and indegrees of all the vertices (extra space can be used if required) in a dense graph of n vertices (number of edges $= \Theta(n^2)$) represented in Adjacency List format is given by $\Theta(n^a \log^b n)$ and $\Theta(n^c \log^d n)$ respectively,
$2 \times a + 3 \times b + 5 \times c + 7 \times d =$ ____

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the DAG G with vertex set $V = \{a, b, c, d, e, f, g\}$ given below. Which of the following is/are a topological ordering of G? (Mark all the correct options)



A. $a\ b\ c\ d\ e\ f\ g$
B. $a\ c\ b\ e\ f\ d\ g$
C. $a\ c\ b\ d\ f\ e\ g$
D. $a\ c\ b\ f\ e\ g\ d$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements regarding topological sorting is/are TRUE? (Mark all the correct options)

A. A topological sort of a dag $G = (V, E)$ is a linear ordering of all its vertices such that if G contains an edge (u, v) then u appears before v in the ordering
B. For better efficiency in performing a topological sort, the graph must be represented using Adjacency List as compared to Adjacency Matrix
C. The worst case time complexity in performing topological sorting of a dag $G = (V, E)$ is $\Theta(|V| + |E|)$
D. If a connected graph is acyclic, it'll have a unique topological sorting

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE? (Mark all the correct options)

A. If in a depth-first traversal of a graph G with n vertices, k edges are marked as tree edges, then the number of connected components in G is $n-k-1$
B. The tightest upper bound on the worst case time complexity in determining the existence of a cycle in an undirected graph $G=(V,E)$ is $\Theta(|V|)$
C. The tightest upper bound on the worst case time complexity in determining the existence of a cycle in an undirected graph $G=(V,E)$ is $\Theta(|V|+|E|)$
D. In a connected graph $G=(V,E)$ with n vertices DFS will yield $n-1$ tree edges

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

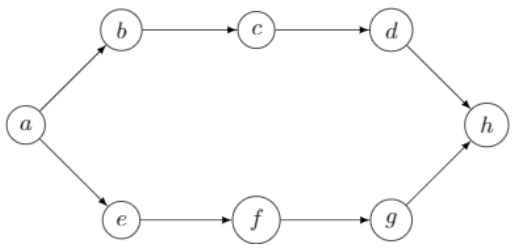
A cut $S=(V-S)$ of an undirected graph $G=(V,E)$ is a partition of V. We say that an edge $(u,v)\in E$ crosses the cut $(S,V-S)$ if one of its endpoints is in S and the other is in $V-S$. An edge is a light edge crossing a cut if its weight is the minimum of any edge crossing the cut. Note that there can be more than one light edge crossing a cut in the case of ties. Mark all the correct options.

A. If an edge is contained in some Minimum Spanning Tree (MST), it is a light edge crossing some cut
B. $\{(u,v) : \text{there exists a cut } (S,V-S) \text{ such that } (u,v) \text{ is a light edge crossing it}\}$ forms a MST
C. A graph G has a unique MST if for every cut, there is a unique light edge crossing the cut
D. If a graph has a unique MST, then for every cut of the graph there is a unique light edge crossing the cut

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following directed graph.



The number of different topological orderings of the vertices of the graph is _____.

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE with respect to the time complexities of the graph algorithms to find Minimum Spanning Trees? (Mark all the appropriate choices)

A. Kruskal's algorithm to find MST runs in $O(|E|\lg|V|)$
B. Prim's algorithm to find MST implemented using binary min-heap runs in $O(|E|\lg|V|)$
C. Prim's algorithm to find MST implemented using Fibonacci heap runs in $O(|E|+|V|\lg|V|)$
D. If graph is given in adjacency list form, Prim's algorithm to find MST runs in $\Theta(V^2)$ using binary min-heap

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE with respect to the time complexities of the graph algorithms to find Single Source Shortest Path (SSSP)? (Mark all the appropriate choices)

A. Bellman-Ford algorithm to find SSSP runs in $O(VE)$
B. Dijkstra algorithm to find SSSP runs in $O(V^2)$ using an array
C. Dijkstra algorithm to find SSSP runs in $O(E \lg V)$ using a binary min-heap
D. SSSP in a Directed Acyclic Graph can be done in $O(|V| + |E|)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 07sec | Discuss

Q #19 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider a complete graph G with 6 vertices. The graph G has _____ spanning trees.

Your Answer: | Correct Answer: 1296 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the graph G with 4 vertices. Its adjacency weight matrix is shown below. Which of the following is true?

$$G = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

A. Graph G has no minimum spanning tree
B. Graph G has a unique minimum spanning trees of cost 3
C. Graph G has 4 distinct minimum spanning trees, each of cost 2
D. Graph M has 16 distinct minimum spanning trees, each of cost 3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

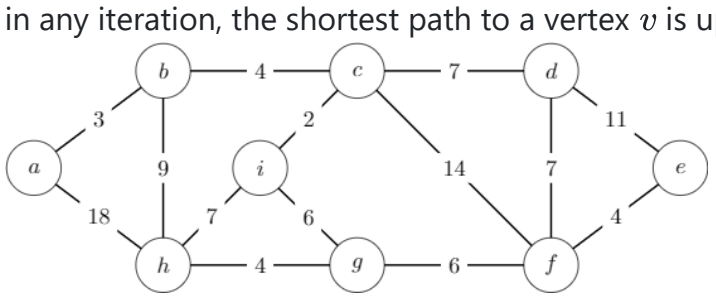
Consider a undirected graph G with n vertices and every edge weight being distinct. Edge e_1 is edge with minimum weight and edge e_2 is edge with maximum weight. Then, which of the following is/are FALSE? (Mark all the applicable options)

A. Every minimum spanning tree of G must contain e_1
B. If e_2 is in a minimum spanning tree, then its removal must disconnect G
C. No minimum spanning tree contains e_2
D. G has a unique minimum spanning tree if e_2 is in a minimum spanning tree

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices c and e. Which one will be reported by Dijstra's shortest path algorithm run with a as the source? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.

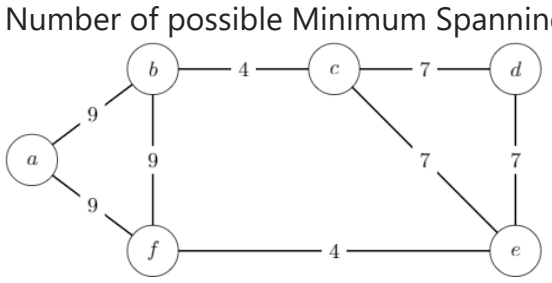


A. $c - d - f - e$
B. $c - f - e$
C. $c - i - g - f - e$
D. $c - d - e$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Number of possible Minimum Spanning Trees for the following graph is _____.



Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

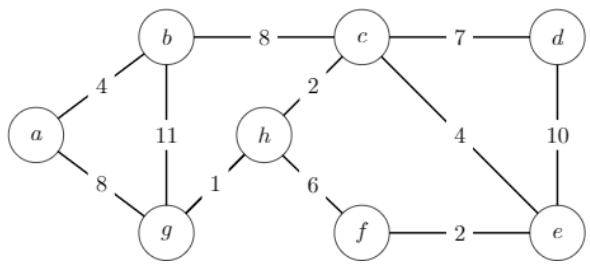
Which of the following is/are correct with respect to Dijkstra's algorithm to find Single Source Shortest Path in a weighted directed graph G? (Mark all the appropriate choices)

A. Dijkstra's algorithm runs in $O(|E| \lg |V|)$ if priority queue is implemented with a binary min-heap
B. For dense graphs ($|E| = \Omega(|V|^2)$) Dijkstra's algorithm performs better if priority queue is implemented with an array
C. Dijkstra's algorithm has a worst case time complexity of $O(|V|^2)$
D. Dijkstra's algorithm has a worst case time complexity of $O(|V| \lg |V| + |E|)$ when implemented with a Fibonacci heap

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose we run Dijkstra's single source shortest-path algorithm on the following edge weighted directed graph with vertex a as the source. In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized? (Assume that whenever there is a tie the vertex whose weight got finalized first is chosen first)



A. a, b, c, h, g, e, d, f
B. a, b, g, h, c, f, e, d
C. a, b, g, c, h, e, d, f
D. a, b, c, h, g, f, e, d

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statement(s) is/are correct regarding Bellman-Ford single source shortest path algorithm?
(Mark all the appropriate choices)

A. Always finds the correct shortest path even when edge weights are negative as long as there is no negative weighted cycle
B. Always finds a negative weighted cycle, if one exists.
C. Finds whether any negative weighted cycle is reachable from the source.
D. Performs better than Dijkstra's algorithm for dense graphs

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Let $G = (V, E)$ be a simple undirected graph with all edge weights being unity, and s be a particular vertex in it called the source. For $x \in V$, let $d(x)$ denote the shortest distance in G from s to x. A breadth first search (BFS) is performed starting at s. Let T be the resultant BFS tree. If (u, v) is an edge of G that is not in T, then the number of possible values of $d(u) - d(v)$ is _____.

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Let G be a directed graph whose vertex set is the set of numbers from 0 to 100. There is an edge from a vertex i to a vertex j if either $j = i + 2$ or $j = 7i$. The minimum number of edges in a path in G from vertex 0 to vertex 100 is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

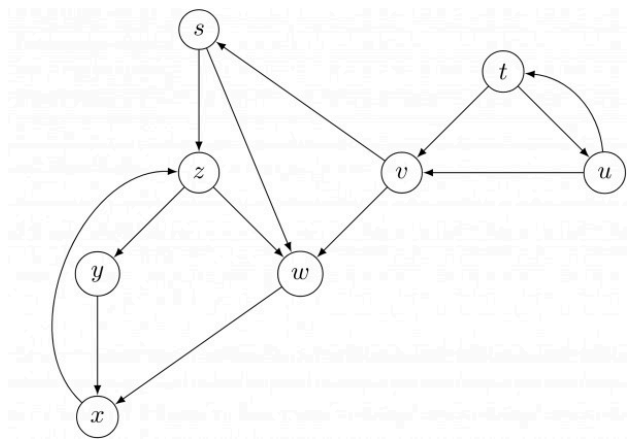
Consider a weighted undirected graph with distinct positive edge weights and let (u, v) be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 36 and the shortest path from s to v has weight 44. Which of the following statements is/are always TRUE? (Mark all the correct options)

A. weight $(u, v) < 8$
B. weight $(u, v) \leq 8$
C. weight $(u, v) > 8$
D. weight $(u, v) \geq 8$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

If s is the source node in the following graph, the number of back edges, forward edges, cross edges and tree edges respectively are (assuming that whenever there is a choice in the depth-first traversal), the vertex earlier in the alphabetical order is to be chosen) _____



A. $3, 2, 2, 6$
B. $2, 4, 1, 6$
C. $3, 2, 1, 4$
D. $2, 1, 4, 6$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff